RISHI THAKUR

DATA SCIENTIST | AI & ADVANCED ANALYTICS

Bengaluru, India • +91 8109830219 • rishi_th219@proton.me • rishi-th219.github.io/portfolio
linkedin.com/in/rishi-thakur-b665a4298 • github.com/rishi-th219

PROFESSIONAL SUMMARY

Data Scientist (M.Tech CSE, GATE CS qualified) experienced in deploying machine learning algorithms, deep learning, and agentic AI systems. Proven track record building predictive models (Random Forest, XGBoost), diffusion pipelines (DDPM/DDIM), and NLP systems (BERT, Transformers). Skilled in AI-assisted prototyping, statistical inference, and large-scale SQL.

TECHNICAL SKILLS

- **Machine Learning & Statistics:** Machine Learning (Random Forest, XGBoost, KMeans, Regression), Probability & Statistics, Hypothesis Testing, Evaluation Metrics (R^2, RMSE, ROC-AUC, F1-Score)
- **GenAI, Agentic Systems & NLP:** Agentic AI (Claude Code, Google Antigravity, Prompt Engineering), NLP (BERT Embeddings, Transformers, Bi-LSTM, 1D-CNN), Text Normalization (NFKD)
- **Deep Learning & Data Systems:** Diffusion Models (DDPM, DDIM), GRU, PyTorch, TensorFlow, Scikit-learn, Resampling (SMOTE + Tomek Links), SQL (CTEs, Window Functions), Power BI

WORK EXPERIENCE

Chhattisgarh Infotech Promotion Society (CHiPS) | Raipur, India | 06/2025 – 03/2026

Data Science & Advanced Analytics Intern • Dept. of Health & Family Welfare, Govt. Of Chhattisgarh

- Collaborated with leadership across **33 districts** to analyze healthcare datasets, delivering models and recommendations.
- Engineered a **3-tier behavioral risk model** (KMeans) on high-volume biometric time-series data to detect attendance leakage, saving public expenditure.
- Authored performant **SQL ETL pipelines** (multi-table CTEs, window functions) bridging biometric punch logs with State Treasury records, resolving discrepancies with **100% data fidelity**.
- Conducted EDA and correlation assessments on multi-source datasets, partnering with NIC engineers to standardize disparate schemas.
- Architected an executive 6-page **Power BI reporting suite** with **4-tier drill-down hierarchies** (District → Facility → Designation → Individual) communicating insights to leadership.

KEY PROJECTS

Proactive Risk Prediction & AI Decision Engine • ***M.Tech Thesis — Deep Generative Modeling & Telemetry***

Technologies: Python, PyTorch, GRU, DDPM/DDIM, Scikit-learn, Random Forest, SMOTE, Tomek Links

- Architected a predictive engine pairing a **GRU encoder** with a **Diffusion Model (DDPM/DDIM)** to forecast signal drops from **spatio-temporal** telemetry.
- Resolved severe **98%/2% class imbalance** using a hybrid **SMOTE + Tomek Links** pipeline, training a two-stage Random Forest classifier to attain a **97.64% F1-score**.

AI vs. Human Text Classification Pipeline • ***Applied NLP, LLM Evaluation & Deep Learning***

Technologies: Python, BERT, XGBoost, Bi-LSTM, Transformers, 1D-CNN, TensorFlow, Scikit-learn

- Constructed dual NLP pipelines extracting 768-dim dense embeddings from pre-trained **BERT** with **XGBoost**, benchmarked against a deep Bi-LSTM/Transformer/1D-CNN network.
- Applied stratified sampling (80/20), text normalization (NFKD), dropout (0.5), and dynamic learning rate decay, achieving **99% Precision** and **98% Recall** in synthetic LLM text detection.

EDUCATION

M.Tech in Computer Science & Engineering | IIIT Naya Raipur (CGPA: 7.90) | 08/2024 – 07/2026

B.Tech in Computer Science | Amity University Chhattisgarh (CGPA: 7.71) | 08/2019 – 06/2023

CERTIFICATIONS & KEY QUALIFICATIONS

- **Qualified GATE Computer Science (CS)**, 2024
- **Python for Data Science, AI & Development** (Coursera)
- **Celonis Foundations** & **Process Management Foundations** (Credly)
- **SQL for Beginners & Database Design** (Udemy)
- **Microsoft Power BI Desktop for Business Intelligence** (Udemy)
- **IELTS Academic:** 7.5 Band Score